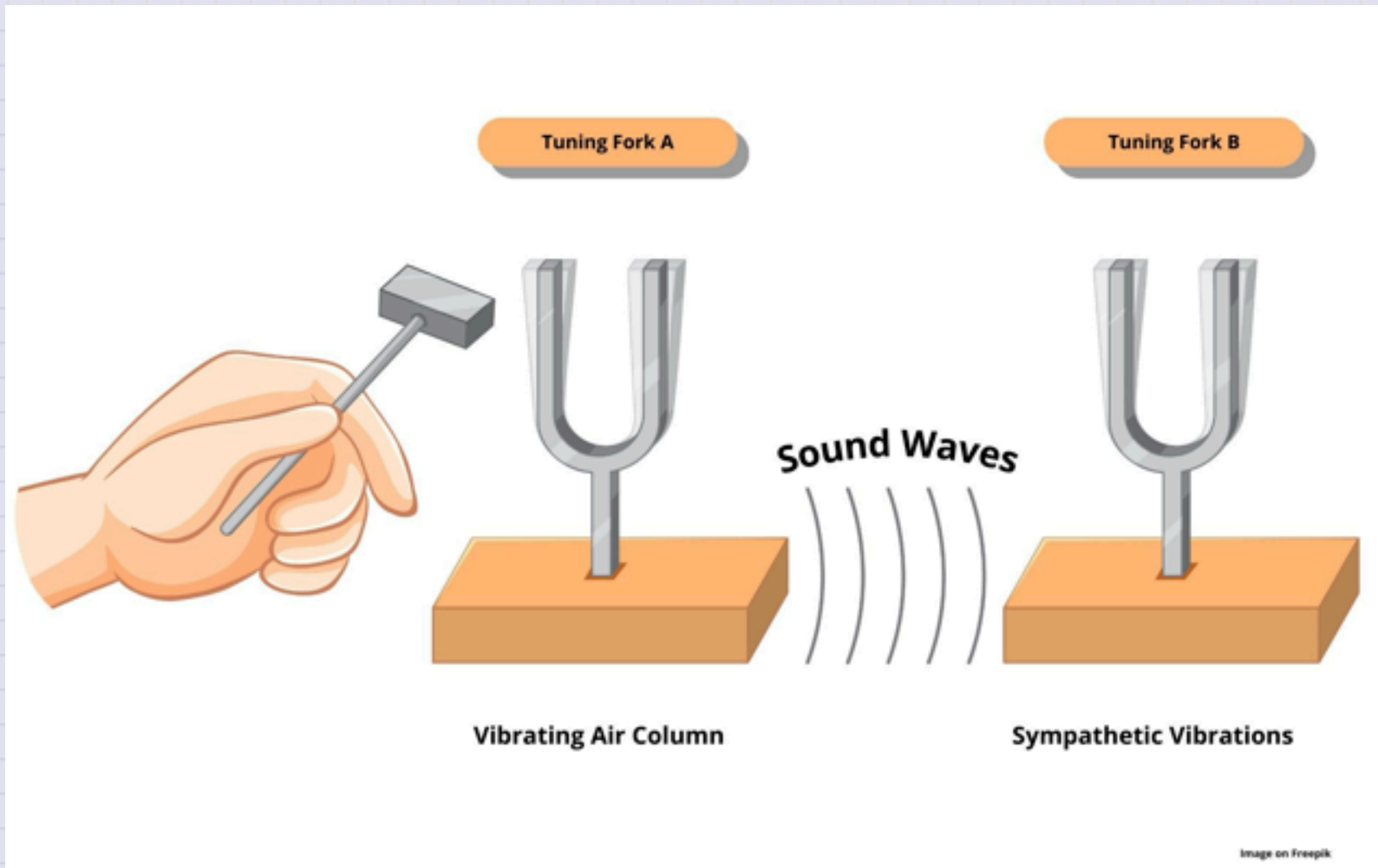


Day 25 - Resonance + Homework Session



Reminders

We started to solve the forced harmonic oscillator equation:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f(t)$$

We examined the case of a sinusoidal driving force:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f_0 \cos(\omega t)$$

There's a complimentary case where the driving force is a sine wave:

$$\ddot{y} + 2\beta\dot{y} + \omega_0^2 y = f_0 \sin(\omega t)$$

Reminders

We combined the two equations into a complex equation using these identities:

$$z(t) = x(t) + iy(t)$$
$$e^{i\omega t} = \cos(\omega t) + i \sin(\omega t)$$

The resulting equation is:

$$\ddot{z} + 2\beta\dot{z} + \omega_0^2 z = f_0 e^{i\omega t}$$

Notice that there's a homogeneous part (z_h) and a particular part (z_p).

$$\ddot{z}_h + 2\beta\dot{z}_h + \omega_0^2 z_h = 0$$

Reminders

The homogeneous part is the solution we've found before with the general solution:

$$z_h(t) = C_1 e^{rt} + C_2 e^{r^*t}$$

where $r = -\beta \pm i\sqrt{\omega_0^2 - \beta^2}$. In the case of a weakly damped oscillator ($\beta^2 < \omega_0^2$), we have:

$$z_h(t) = e^{-\beta t} \left(C_1 e^{-i\sqrt{\omega_0^2 - \beta^2}t} + C_2 e^{+i\sqrt{\omega_0^2 - \beta^2}t} \right)$$

These solutions die out as $t \rightarrow \infty$. They are called **transient solutions**.

Solving the particular part

The particular part is the solution to the driven harmonic oscillator equation:

$$\ddot{z}_p + 2\beta\dot{z}_p + \omega_0^2 z_p = f_0 e^{i\omega t}$$

Assume a sinusoidal solution (frequency, ω) of the form:

$$z_p(t) = C e^{i\omega t}$$

where C is a complex number. Then, we have:

$$\begin{aligned} -\omega^2 C e^{i\omega t} + 2i\beta\omega C e^{i\omega t} + \omega_0^2 C e^{i\omega t} &= f_0 e^{i\omega t} \\ (-\omega^2 + 2i\beta\omega + \omega_0^2) C &= f_0 \end{aligned}$$

Amplitude of the particular solution

$$C = \frac{f_0}{(\omega_0^2 - \omega^2 + 2i\beta\omega)}$$

We want to convert this to polar form:

$$C = Ae^{-i\delta}$$

where A and δ are real numbers. We use the complex form to compute the magnitude of the amplitude:

$$A^2 = C\bar{C} = \frac{f_0^2}{(\omega_0^2 - \omega^2 + 2i\beta\omega)(\omega_0^2 - \omega^2 - 2i\beta\omega)}$$

$$A^2 = \frac{f_0^2}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}$$

Clicker Question 24-5

We found that the square amplitude of the driven harmonic oscillator is:

$$A^2 = \frac{f_0^2}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}$$

When is the amplitude of the driven oscillator maximized?

1. When the driving frequency (ω) is far from the natural frequency (ω_0)
2. When the driving frequency (ω) is close to the natural frequency (ω_0)
3. When the damping (2β) is weak
4. When the damping (2β) is strong
5. Some combination of the above

Finding the phase

With,

$$C = \frac{f_0}{(\omega_0^2 - \omega^2 + 2i\beta\omega)} = Ae^{-i\delta}$$

then we can compare the complex forms:

$$f_0 e^{i\delta} = A (\omega_0^2 - \omega^2 + 2i\beta\omega).$$

Both f_0 and A are real numbers, so the phase δ is the same phase as the complex number:

$$\delta = \tan^{-1} \left(\frac{2\beta\omega}{\omega_0^2 - \omega^2} \right)$$

The Particular Solution

Let's return to the particular solution:

$$z_p(t) = Ce^{i\omega t} = Ae^{-i\delta}e^{i\omega t} = Ae^{i(\omega t - \delta)}$$
$$z_p(t) = A \cos(\omega t - \delta) + iA \sin(\omega t - \delta)$$

So we get solutions to both driven oscillators:

$$x_p(t) = \operatorname{Re}(z_p(t)) = A \cos(\omega t - \delta)$$
$$y_p(t) = \operatorname{Im}(z_p(t)) = A \sin(\omega t - \delta)$$

These are the **steady-state solutions**.

They persist as $t \rightarrow \infty$ and oscillate at the driving frequency ω .

The Full Solution

$$x(t) = x_h(t) + x_p(t)$$

Here, $x_h(t)$ is the transient solution and $x_p(t)$ is the steady-state solution.

For weakly damped oscillators, the transient solution can be written in the form:

$$x_h(t) = A_{tr} e^{-\beta t} \cos(\sqrt{\omega_0^2 - \beta^2} t + \delta_{tr})$$

where A_{tr} and δ_{tr} are real numbers and are the amplitude and phase of the transient solution. Both are determined by the initial conditions.

$$x_p(t) = A \cos(\omega t - \delta)$$

where A and δ are real numbers and are the amplitude and phase of the steady-state solution.

The Full Solution

The transient plus the steady-state solution is the full solution:

$$x(t) = A_{tr}e^{-\beta t} \cos(\sqrt{\omega_0^2 - \beta^2}t + \delta_{tr}) + A \cos(\omega t - \delta)$$

As $t \rightarrow \infty$, the transient solution dies out and the steady-state solution persists.

$$x(t \rightarrow \infty) = A \cos(\omega t - \delta)$$

where

$$A = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}}$$

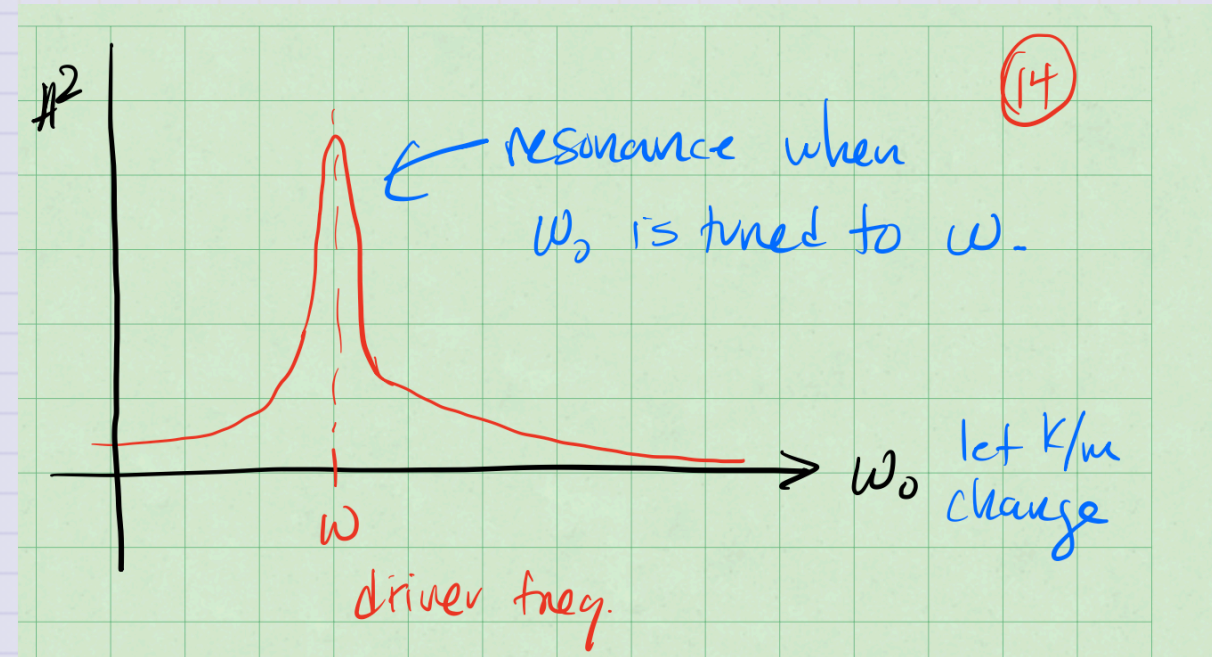
$$\delta = \tan^{-1} \left(\frac{2\beta\omega}{\omega_0^2 - \omega^2} \right)$$

Resonance

The amplitude of the steady-state solution is:

$$A^2 = \frac{f_0^2}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}$$

We change ω_0 and observe how the amplitude changes.



Achieving resonance

The denominator of the equation controls the amplitude:

$$(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2$$

Case 1: Tune ω_0 to be close to ω . *Car Radio tuning*

With $\omega_0 = \omega$, the amplitude is:

$$A^2 = \frac{f_0^2}{4\beta^2\omega^2}$$

Achieving resonance

The denominator of the equation controls the amplitude:

$$(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2$$

Case 2: Tune ω to be close to ω_0 . *Pushing a swing*

Find the ω that maximizes the amplitude by taking the derivative with respect to ω :

$$\frac{d}{d\omega} ((\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2) = 0$$

$$2(\omega_0^2 - \omega^2)(-2\omega) + 8\beta^2\omega = 0$$

$$4\omega(\omega^2 - \omega_0^2 + 2\beta^2) = 0$$

$$\omega = 0 \quad \omega = \sqrt{\omega_0^2 - 2\beta^2}$$

HW6 Exercise 2 Morse Potential as an SHO

If the potential has a local minimum, we can often find SHO approximation for that potential near the local minimum.

The **Morse potential** is a convenient model for the potential energy of a diatomic molecule. The potential is a radial one and thus one-dimensional. It is given by,

$$U(r) = A \left[\left(e^{(R-r)/S} - 1 \right)^2 - 1 \right]$$

where the distance between the centers of the two atoms is r , and the constants A , R , and S are all positive. Here $S \ll R$.

- 2a (2pt) Sketch (or plot) the potential as a function of r .

HW6 Exercise 2 Morse Potential as an SHO

$$U(r) = A \left[\left(e^{(R-r)/S} - 1 \right)^2 - 1 \right]$$

- 2b (3pt) Find the equilibrium position of the potential, i.e. the position where the potential is at a minimum. We will call this r_e .
- 2c (3pt) Rewrite the potential in terms of the displacement from equilibrium, $r = r_e + x$. Expand the potential to second order in x .
- 2d (2pt) Find the effective spring constant, k , for the potential near the minimum. What is the frequency of small oscillations about the minimum?

HW6 Exercise 4, Toy Potential

Consider a toy potential of the form,

$$U(r) = U_0 \left(\frac{r}{R} + \lambda^2 \frac{R}{r} \right)$$

where U_0 , R , and λ are all positive constants and the domain of the potential is $0 < r < \infty$.

- 4a (2pt) Sketch (or plot) the potential as a function of r .

HW6 Exercise 4, Toy Potential

$$U(r) = U_0 \left(\frac{r}{R} + \lambda^2 \frac{R}{r} \right)$$

- 4b (3pt) Find the equilibrium position of the potential, i.e. the position where the potential is at a minimum. We will call this r_e .
- 4c (5pt) Rewrite the potential in terms of the displacement from equilibrium, $r = r_e + x$. Expand the potential to second order in x . What is the effective spring constant, k , for the potential near the minimum? What is the frequency of small oscillations about the minimum?